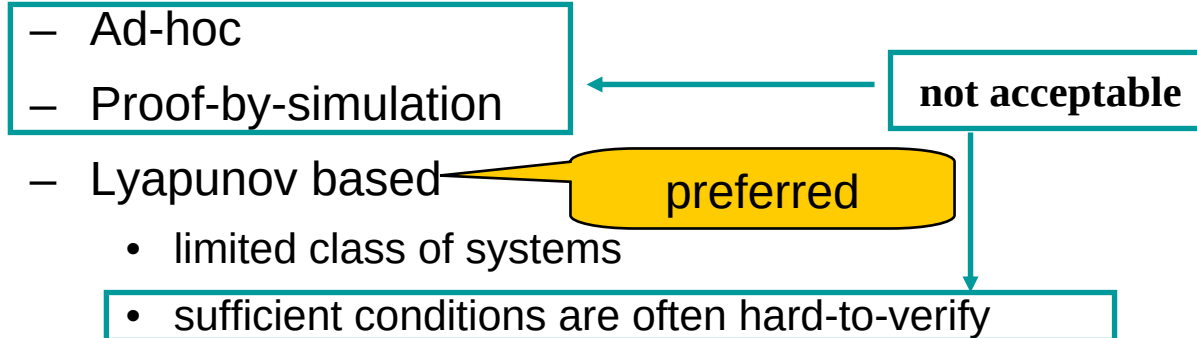


Adaptive Control in the Presence of Actuator Constraints

Overview

- Problem: Assure stability of an adaptive control system in the presence of actuator position / rate saturation constraints.

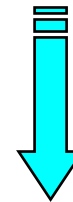
- Solutions



- Need: *Theoretically justified* and *verifiable* conditions for stable adaptation and control design with a possibility of avoiding actuator saturation phenomenon.
- Design Solutions include modifications, (adaptive / fixed gain) to:
 - control input
 - tracking error
 - reference model

Known Design Solutions

- R. Monopoli, (1975)
 - adaptive modifications: tracking error and reference input
 - no theoretical stability proof
- S.P. Karason, A.M. Annaswamy, (1994)
 - adaptive modifications: reference input
 - rigorous stability proof
- E.N. Johnson, A.J. Calise, (2003)
 - pseudo control hedging (PCH)
 - fixed gain modification of reference input
- E. Lavretsky, N. Hovakimyan, (2004)
 - positive μ – modification
 - adaptive modification of control and reference inputs
 - rigorous stability proof and verifiable sufficient conditions
 - capability to completely avoid control saturation



Adaptive Control in the Presence of Input Constraints: Problem Formulation

- System dynamics: $\dot{x}(t) = Ax(t) + b\lambda u(t), \quad x \in R^n, u \in R$
 - A is unknown matrix,
 - b is known control direction
 - $\lambda > 0$ unknown positive constant

- Static actuator

$$u(t) = u_{\max} \operatorname{sat}\left(\frac{u_c}{u_{\max}}\right) = \begin{cases} u_c(t), & |u_c(t)| \leq u_{\max} \\ u_{\max} \operatorname{sgn}(u_c(t)), & |u_c(t)| \geq u_{\max} \end{cases}$$

amplitude saturation

- Ideal Reference model dynamics:

$$\dot{x}_m^*(t) = A_m x_m^*(t) + b_m r(t), \quad x_m^* \in R^n, r \in R$$

commanded input

Hurwitz

bounded reference input

Preliminaries

- Define: $u_{\max}^{\delta} = u_{\max} - \delta$, where: $0 < \delta < u_{\max}$

safety zone
- Commanded control deficiency: $\rightarrow \Delta u_c = u_{\max}^{\delta} \operatorname{sat}\left(\frac{u_c}{u_{\max}^{\delta}}\right) - u_c$
- Adaptive control with μ -modification, (*implicit* form):

$$u_c = \underbrace{k_x^T x + k_r r}_{u_{lin}} + \boxed{\mu \Delta u_c}$$

linear feedback / feedforward component

control deficiency feedback
- Need *explicit* form of u_c

Positive μ – modification

- Adaptive control with μ – mod is given by convex combination of u_{lin} and $u_{\max}^{\delta} \text{sat}\left(\frac{u_{lin}}{u_{\max}^{\delta}}\right)$

$$u_c = \frac{1}{1+\mu} \left(u_{lin} + \mu u_{\max}^{\delta} \text{sat}\left(\frac{u_{lin}}{u_{\max}^{\delta}}\right) \right) = \begin{cases} u_{lin}, & |u_{lin}| \leq u_{\max}^{\delta} \\ \frac{1}{1+\mu} (u_{lin} + \mu u_{\max}^{\delta}), & u_{lin} > u_{\max}^{\delta} \\ \frac{1}{1+\mu} (u_{lin} - \mu u_{\max}^{\delta}), & u_{lin} < -u_{\max}^{\delta} \end{cases}$$

continuous in time but not continuously differentiable

$$u_c = u_{lin} + \mu \Delta u_c$$

$$\Delta u_c = \left(u_{\max}^{\delta} \text{sat}\left(\frac{u_{lin}}{u_{\max}^{\delta}}\right) - u_{lin} \right) \frac{1}{1+\mu}$$

Closed-Loop Dynamics

- μ – mod control: $u_c = u_{lin} + \mu \Delta u_c$
- System dynamics: $\dot{x} = Ax + b\lambda u_c + b\lambda \overbrace{(u - u_c)}^{\Delta u}$
- Closed-loop system:

$$\dot{x} = Ax + b\lambda u_{lin} + b\lambda \overbrace{(\mu \Delta u_c + \Delta u)}^{\Delta u_{lin} = u - u_{lin}}$$

where: $\Delta u_{lin} = u_{max} \operatorname{sat}\left(\frac{u_c}{u_{max}}\right) - u_{lin}$

linear control deficiency



does not depend on μ explicitly

$$\dot{x} = (A + b\lambda k_x^T)x + b\lambda(k_r r + \Delta u_{lin})$$

Adaptive Reference Model Modification

- Closed-loop system:

$$\dot{x} = (A + b \lambda k_x^T) x + b \lambda (k_r r + \Delta u_{lin})$$

- Leads to consideration of adaptive reference model:

$$\dot{x}_m = A_m x_m + b_m \left(r(t) + \boxed{k_u \Delta u_{lin}} \right), \quad |r(t)| \leq r_{\max}$$

adaptive augmentation

reference input

- Matching conditions:

$$\forall \lambda > 0 \exists (k_x^* \in R^n, \quad k_r^* \in R, \quad k_u^* \in R)$$



$$\begin{cases} A + b \lambda (k_x^T)^* = A_m \\ b \lambda k_r^* = b_m \\ b \lambda = b_m k_u^* \end{cases} \Rightarrow k_u^* k_r^* = 1$$

Adaptive Laws Derivation

- Tracking error: $e = x - x_m$
- Parameter errors:  $\begin{cases} \Delta k_x = k_x - k_x^* \\ \Delta k_r = k_r - k_r^* \\ \Delta k_u = k_u - k_u^* \end{cases}$

- Tracking error dynamics:

$$\dot{e} = A_m e + b \lambda \left(\Delta k_x^T x + \Delta k_r r \right) - b_m \Delta k_u \Delta u_{lin}$$

- Lyapunov function:

$$V(e, \Delta k_x, \Delta k_r, \Delta k_u) = e^T P e + \lambda \left(\Delta k_x^T \Gamma_x^{-1} \Delta k_x + \gamma_r^{-1} \Delta k_r^2 + \gamma_u^{-1} \Delta k_u^2 \right)$$

$$\text{where: } P A_m + A_m^T P = -Q < 0$$

Stable Parameter Adaptation

- Adaptive laws derived to yield stability:

$$\begin{cases} \dot{k}_x = -\Gamma_x x e^T P b \\ \dot{k}_r = -\gamma_r r(t) e^T P b \\ \dot{k}_u = \gamma_u \Delta u_{lin} e^T P b_m \end{cases} \Leftrightarrow \boxed{\dot{V} = -e^T Q e < 0} \Rightarrow \boxed{\dot{V}(e, \Delta k_x, \Delta k_r, \Delta k_u) \leq 0}$$

- For open-loop stable systems – global result
- For open-loop unstable systems verifiable sufficient conditions established:
 - upper bound on $r_{\max}$
 - lower bound on μ
 - upper bounds on initial conditions $x(0)$ and Lyapunov function $V(0)$

μ – mod Design Steps

- Choose “safety zone” $0 < \delta < u_{\max}$ and sufficiently large $\mu > 0$
- Define *virtual* constraint: $u_{\max}^{\delta} = u_{\max} - \delta$
- Linear* component of adaptive control signal: $u_{lin} = k_x^T x + k_r r(t)$
- Total adaptive control with μ – mod:

$$u_c = \frac{1}{1 + \mu} \left(u_{lin} + \mu u_{\max}^{\delta} \text{sat} \left(\frac{u_{lin}}{u_{\max}^{\delta}} \right) \right)$$

adaptive laws

$$\begin{cases} \dot{k}_x = -\Gamma_x x e^T P b \\ \dot{k}_r = -\gamma_r r(t) e^T P b \\ \dot{k}_u = \gamma_u \Delta u_{lin} e^T P b_m \end{cases}$$

$$\dot{x}_m = A_m x_m + b_m \left(r + k_u \underbrace{\left(u_{\max} \text{sat} \left(\frac{u_c}{u_{\max}} \right) - u_{lin} \right)}_{\Delta u_{lin}} \right)$$

modified reference model

Simulation Example

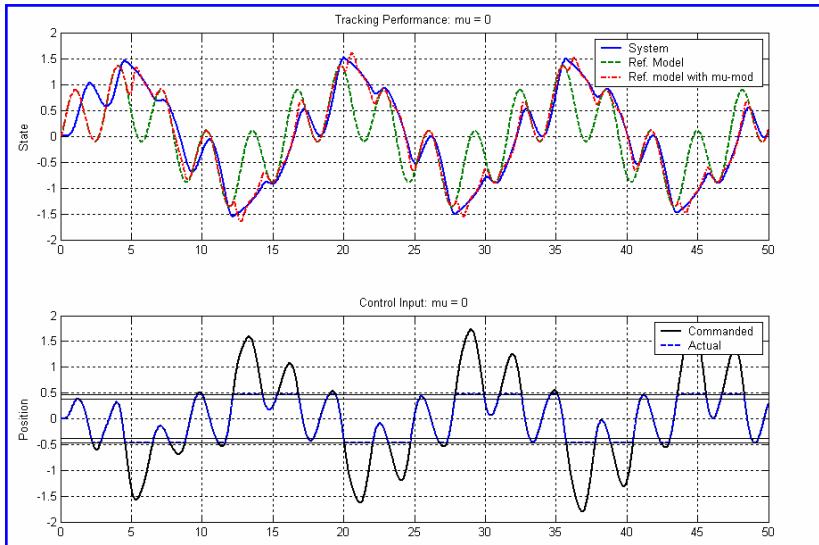
- Unstable open-loop system:

$$\dot{x} = a x + b u_{\max} \operatorname{sat}\left(\frac{u_c}{u_{\max}}\right), \text{ where: } a = 0.5, b = 2, u_{\max} = 0.47$$

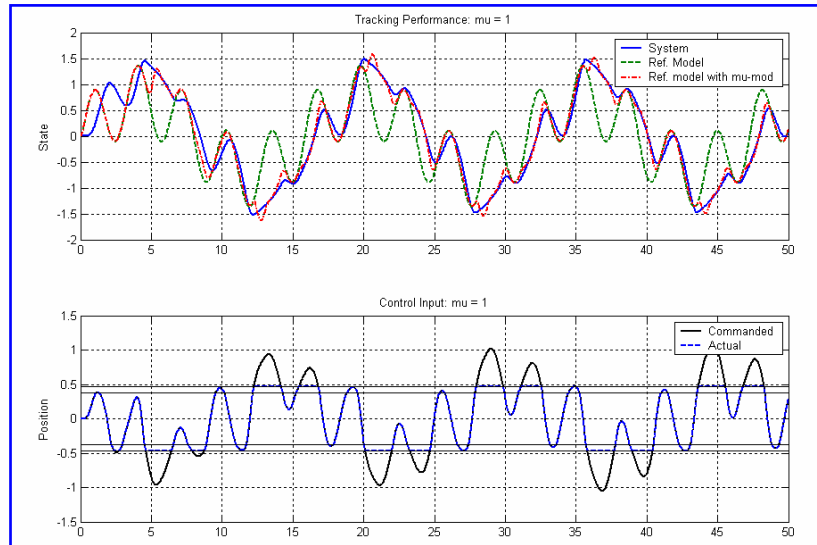
- Choose: $\delta = 0.2 u_{\max}$ $\longrightarrow$ $u_{\max}^{\delta} = u_{\max} - \delta = 0.8 u_{\max}$
- Ideal reference model: $\dot{x}_m = -6(x_m - r(t))$
- Reference input: $r(t) = 0.7(\sin(2t) + \sin(0.4t))$
- Adaptation rates set to unity
- System and reference model start at zero

Simulation Data

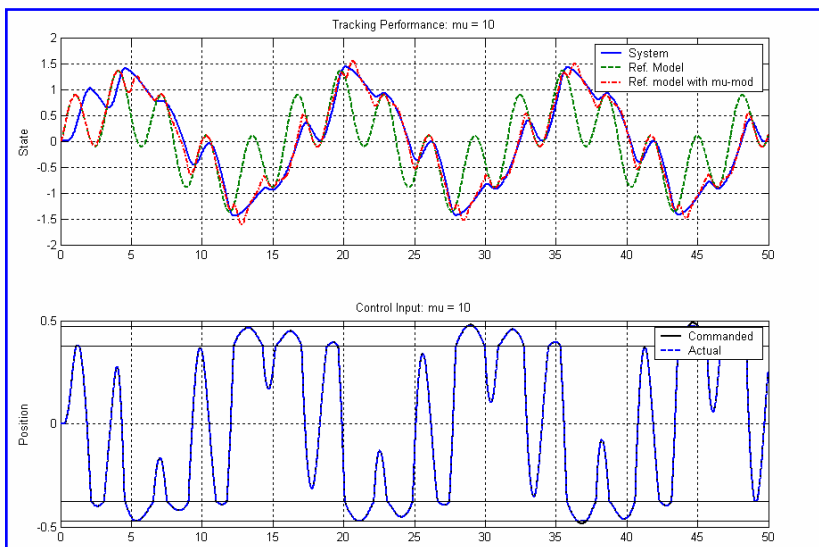
$\mu = 0$



$\mu = 1$



$\mu = 10$



$\mu = 100$

